

Introduction Background

- Traditional credit scoring models (e.g. FICO Score) often fail to account for individuals lacking a conventional credit history
- Our project leverages alternative financial data, such as categorized bank transactions and income predictions, to develop a fairer creditworthiness assessment with advanced machine learning (ML) models

Research Question

- How can ML be applied to develop a "Cash Score" that accurately reflects financial behavior and equal access to credit? Can we predict if someone will credit default or not based on their behavior?

Data Overview

- Bank transaction data (2017-2023) provided by PrismData. Includes categorized inflow and outflow transactions sufficient to extract features like: income levels, spending habits, balance changes, and other consumer-level financial behaviors.

Feature Engineering

We created hundreds of features based on attributes in our datasets. Our features fall under 3 types concerned with:

- Bank Balance Features:** Measure account balance trends, including current balance, balance changes over time, and average account balance.
- Income Features:**Capture income-related metrics like average transaction amounts over different time frames, net monthly cash flow, and detailed statistics on transaction categories.
- Spending Features:** Analyze spending patterns through outflow statistics over different time periods, including yearly, monthly, and weekly trends.

(Include a feature plot here to go in-depth on feature creation)

While developing these features, we had to ensure our model remained unbiased. In the financial services industry, compliance with the Equal Credit Opportunity Act (ECOA) is essential. This meant removing certain features—not only based on their impact on model performance but also to prevent unintentional bias toward specific demographics.

Feature Selection

- Feature importance measured using SHapley Additive exPlanations (SHAP)

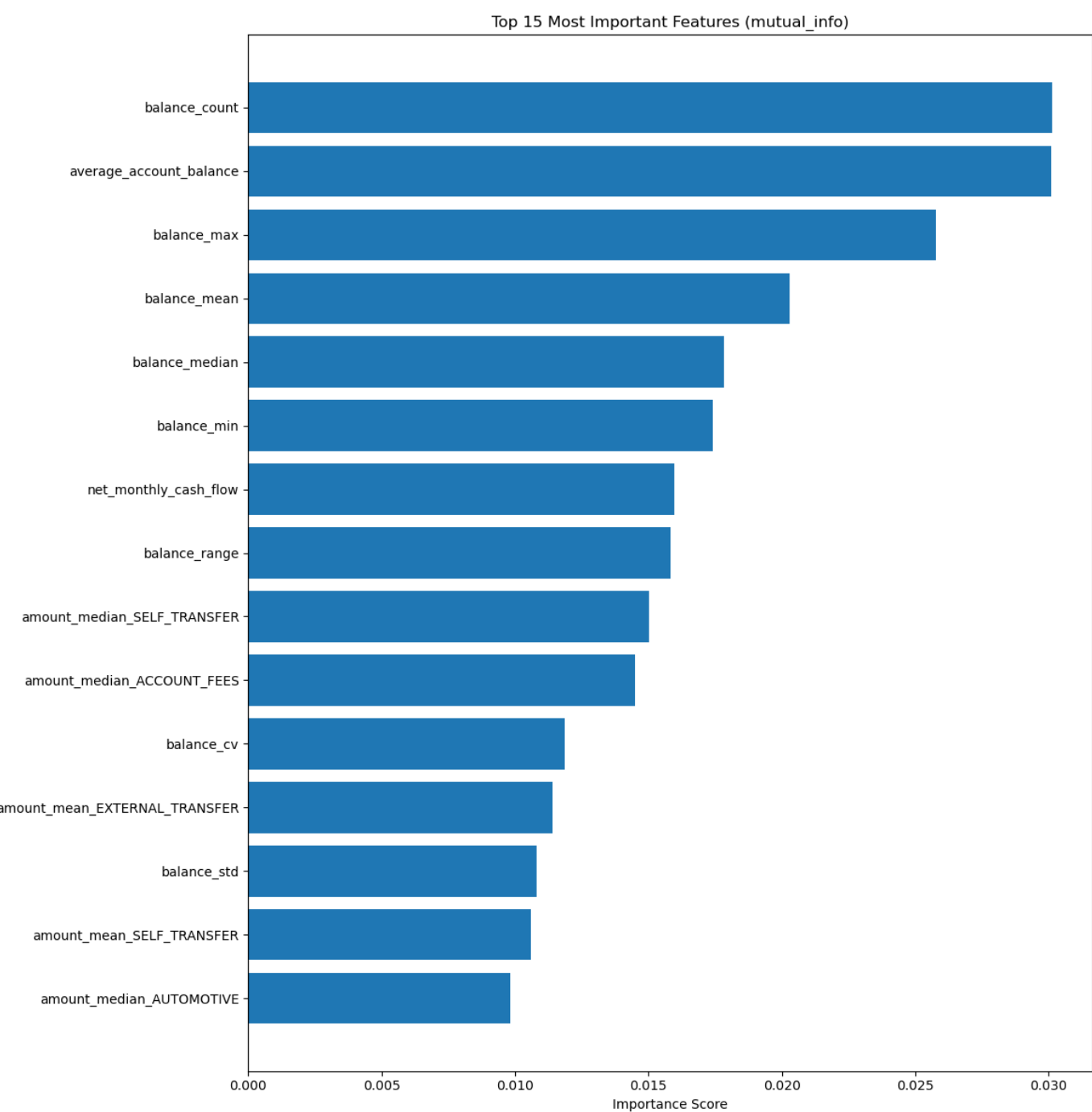


Figure 1. Top 10 Features Selected by Mutual Information Metric

Machine Learning Models

- **Baseline:** Logistic Regression
- **Best Model:** XGBoost (optimized with hyperparameter tuning)

Reason Codes

- (Explain Shap score process for possible score exclusions in future)

Evaluating Our Scores Against the FICO Score

- (Table evaluating cash score versus traditional credit score models)

Results

- **Best Model: XGBClassifier**
- **Accuracy: **
- **ROC-AUC Score: **

Feature Selection	Model	Test Accuracy	Test Balanced Accuracy	Test ROC AUC
all_features	xgboost	0.78	0.71	0.79
all_features	gradient_boosting	0.92	0.51	0.81
all_features	random_forest	0.92	0.52	0.78
all_features	logistic	0.67	0.68	0.71
mutual_info	xgboost	0.75	0.71	0.8
mutual_info	gradient_boosting	0.92	0.53	0.8
mutual_info	random_forest	0.92	0.52	0.79
mutual_info	logistic	0.63	0.69	0.73
random_forest	xgboost	0.76	0.72	0.79
random_forest	gradient_boosting	0.92	0.54	0.81
random_forest	random_forest	0.92	0.52	0.77
random_forest	logistic	0.63	0.7	0.72
xgboost	xgboost	0.75	0.72	0.8
xgboost	gradient_boosting	0.92	0.51	0.8
xgboost	random_forest	0.92	0.52	0.79
xgboost	logistic	0.64	0.69	0.71

Figure 2. XGBoost Model Performance on Test and Training Sets

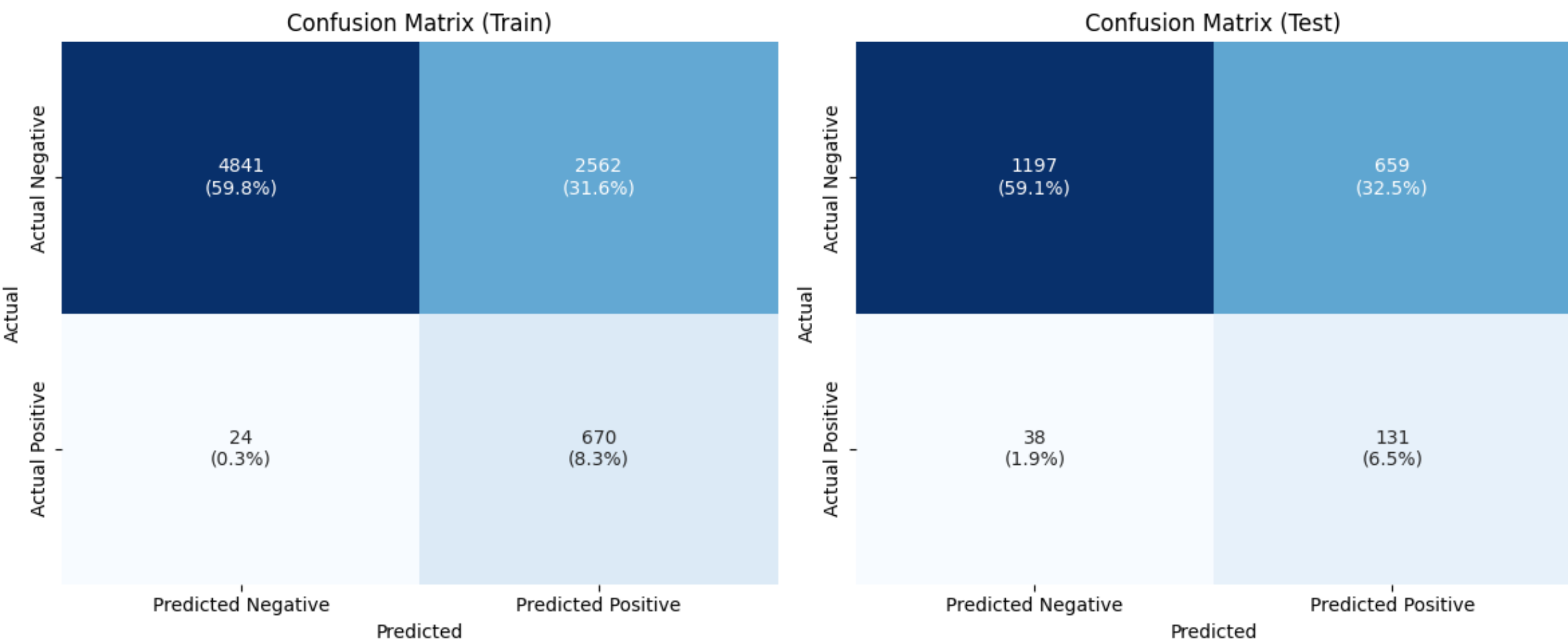


Figure 3. Confusion Matrices for Test and Training Sets

Conclusion

- Our "Cash Score" provides a more inclusive credit evaluation.
- Real-time transaction data enhances creditworthiness assessment.
- Future Work:

- Ensure fairness, reduce bias.
- Optimize ML algorithms